Bayesian Hierarchical Logistic Regression for 30-Day Hospital Readmission Among Diabetic Patients: The Role of Medical Specialty

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Abstract

Hospital readmission within 30 days is a costly and clinically important outcome. We investigate whether medical specialty contributes to readmission risk after adjusting for patient characteristics using a Bayesian hierarchical logistic regression with specialty-specific random intercepts, fitted via a custom Metropolis-within-Gibbs sampler on 37,964 diabetic encounters across 19 specialties. Substantial between-specialty heterogeneity is observed ($\hat{\tau} = 0.437$, 95% CrI $[0.298, 0.644]$), with Physical Medicine and Rehabilitation exhibiting markedly elevated risk. Key predictors include prior inpatient visits (OR = 1.22), comorbid neoplasms (OR = 1.51), and A1C measurement (OR = 1.42). Results are robust to prior sensitivity analysis.

1 Introduction

Unplanned hospital readmissions within 30 days of discharge represent a major burden on healthcare systems, with annual costs exceeding $26 billion in the United States. Patients with diabetes, characterised by high comorbidity and complex medication regimens, experience particularly high readmission rates. The Centers for Medicare & Medicaid Services (CMS) has introduced financial penalties for hospitals with excess readmissions, making reduction efforts both clinically and economically important.

Previous studies have identified patient-level predictors of readmission, such as length of stay, prior admissions, and comorbidity burden, typically using logistic regression or machine learning models. However, an important structural factor is often overlooked: patients are managed by physicians from different medical specialties, which may vary in post-discharge care patterns, case mix, and practice style. Ignoring this clustering can violate independence assumptions, bias standard errors, and obscure opportunities for specialty-level intervention.

This study examines whether specialty-level heterogeneity contributes to 30-day readmission risk among diabetic patients after adjusting for patient-level covariates. The analysis is complicated by substantial group-size imbalance, with some specialties contributing large numbers of encounters while others have relatively few, making naïve comparisons unreliable due to sampling variability.

A Bayesian hierarchical model is well suited to this setting. Partial pooling allows small-sample specialties to borrow strength from larger groups through the shared hyperparameter τ, yielding more stable estimates than complete or no pooling. The Bayesian framework provides coherent uncertainty quantification via posterior credible intervals for τ, directly characterising between-specialty variation without reliance on p-values. Model assessment can be conducted using DIC and posterior predictive checks within a unified framework. In addition, the hierarchical prior induces shrinkage on random intercepts, acting as adaptive regularisation and controlling model complexity without requiring cross-validation.

2 Data and Cohort Definition

2.1 Data Source

We use the Diabetes 130-US Hospitals dataset [5], comprising 101,766 inpatient encounters from 130 US hospitals over 1999-2008, with detailed demographic, clinical, and medication information.

2.2 Cohort Construction

Starting from $N_{\text{raw}} = 101{,}766$, variables with high missingness or no variation were removed, including weight, payer$_c ode, citoglipton, andexamide.Toensureindependence, onlythefirstencounterperpatientwasretained, yieldingN$ $= 71{,}518.Encounterscorrespondingtoexpiredorhospicedischargeswereexcluded, leavingN = 69{,}973.Theoutcomewasdefineda$

1 for readmission within 30 days (< 30), and 0 otherwise. Encounters with unknown specialty were excluded (45.7%), resulting in $N = 37{,}965$, and specialties were grouped into $J = 19$ categories by merging small groups. One record with unknown gender was removed, yielding a final sample size of $N = 37{,}964$. Continuous variables were standardised to z-scores, and categorical features were constructed from demographics, ICD-9 diagnosis groups, comorbidities, and medication indicators. The final dataset includes $N = 37{,}964$ encounters, $J = 19$ specialties, $K = 43$ covariates, and a 30-day readmission rate of 9.60%.

The cohort has a mean age of 64.9 years, with 53.5% female and 73.0% Caucasian patients. Readmitted patients are older and exhibit greater clinical complexity, including longer stays, more medications, more prior inpatient visits, and more diagnoses.

Specialty group sizes range from 288 (Physical Medicine and Rehabilitation) to 10,975 (Internal Medicine), with raw readmission rates spanning 4.3% (Obstetrics/Gynecology) to 32.3% (Physical Medicine and Rehabilitation). The wide dispersion in unadjusted readmission rates motivates a model that accounts for specialty-level heterogeneity.

3 Model Specification

3.1 Notation

Table 1: Summary of notation.

Symbol	Meaning
$N = 37{,}964$	Number of observations (encounters)
$K = 43$	Number of patient-level covariates
$J = 19$	Number of medical specialty groups
$y_i \in \{0, 1\}$	30-day readmission indicator for encounter i
$\boldsymbol{x}_i \in \mathbb{R}^K$	Covariate vector for encounter i
$j[i] \in \{1, \ldots, J\}$	Specialty group index for encounter i
n_j	Number of observations in group j
$\alpha \in \mathbb{R}$	Global intercept
$\boldsymbol{\beta} \in \mathbb{R}^K$	Fixed-effect coefficients
$u_j \in \mathbb{R}$	Random intercept for specialty group j
$\boldsymbol{u} = (u_1, \ldots, u_J)^\top$	Vector of all random intercepts
$\tau > 0$	Between-group standard deviation
η_i	Linear predictor for observation i

3.2 Likelihood

Let $y_i \in \{0, 1\}$ denote whether encounter i resulted in 30-day readmission. Conditional on parameters, outcomes are independent Bernoulli:

$$y_i \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u} \overset{\text{ind}}{\sim} \text{Bernoulli}(\pi_i), \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad \eta_i = \alpha + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_{j[i]}, \tag{1}$$

where $\boldsymbol{x}_i \in \mathbb{R}^K$ is the covariate vector ($K = 43$), $j[i] \in \{1, \ldots, J\}$ indicates the specialty group, and $u_{j[i]}$ is the specialty-specific random intercept.

The full likelihood is:

$$p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}) = \prod_{i=1}^{N} \frac{\exp(y_i\, \eta_i)}{1 + \exp(\eta_i)}. \tag{2}$$

3.3 Prior Distributions

$$\alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \qquad \sigma_\alpha = 5, \tag{3}$$

$$\beta_k \overset{\text{iid}}{\sim} \mathcal{N}(0, s_\beta^2), \qquad s_\beta = 2.5, \quad k = 1, \ldots, K, \tag{4}$$

$$u_j \mid \tau \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \qquad j = 1, \ldots, J, \tag{5}$$

$$\tau \sim \text{Half-Normal}(0, s_\tau^2), \qquad s_\tau = 1.0. \tag{6}$$

The Half-Normal density is $p(\tau) = \frac{2}{\sqrt{2\pi}\, s_\tau} \exp\!\left(-\frac{\tau^2}{2 s_\tau^2}\right)$ for $\tau > 0$. As a sensitivity alternative (Section 8), we also consider $\tau \sim \text{Half-Cauchy}(0,\, s_\tau)$ with density $p(\tau) = \frac{2}{\pi\, s_\tau} \frac{1}{1 + \tau^2 / s_\tau^2}$, $\tau > 0$.

The priors are weakly informative [7], chosen to impose minimal substantive constraint while ensuring posterior propriety:

- **Intercept** ($\sigma_\alpha = 5$): on the logit scale, $\alpha \in [-10, 10]$ covers baseline probabilities from $\approx 0.005\%$ to $\approx 99.995\%$, far wider than any clinically plausible readmission rate.
- **Fixed effects** ($s_\beta = 2.5$): allows odds ratios up to $\sim e^5 \approx 150$; in practice, patient-level ORs for readmission rarely exceed 3–5, so this prior is nearly non-informative for well-identified coefficients while providing gentle regularisation for weakly identified ones.
- **Hyperprior on** τ: Half-Normal$(0, 1)$ concentrates mass near zero while permitting substantial heterogeneity [3]. This choice reflects the prior belief that specialty effects, while potentially present, are unlikely to dominate patient-level predictors. On the log-odds scale, $\tau = 1$ corresponds to a roughly 2.7-fold variation in odds between specialties at ± 1 SD, which is already large.

3.4 Hierarchical Structure

The key feature is the exchangeable random intercepts $u_j \mid \tau \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau^2)$. The shared scale τ induces partial pooling: specialty estimates are shrunk toward the global mean, with the degree of shrinkage inversely related to group size n_j.

The posterior mean of u_j can be understood as a compromise between two extremes. In the normal-approximation limit, the Bayes estimate approximately equals:

$$\hat{u}_j^{\text{Bayes}} \approx \underbrace{\frac{n_j/\hat{\sigma}_j^2}{n_j/\hat{\sigma}_j^2 + 1/\hat{\tau}^2}}_{\text{shrinkage weight } \lambda_j} \hat{u}_j^{\text{MLE}}, \tag{7}$$

where $\hat{\sigma}_j^2$ is the within-group variance of the linear predictor residuals. The shrinkage weight $\lambda_j \in [0, 1]$ depends on the signal-to-noise ratio within each group:

- When $\tau \to 0$, $\lambda_j \to 0$ for all j: all specialties collapse to a single intercept (**complete pooling**).
- As $\tau \to \infty$, $\lambda_j \to 1$ for all j: each group estimates its own intercept independently (**no pooling**).
- For intermediate τ (the empirically relevant case), large groups (n_j large, e.g., Internal Medicine with $n = 10{,}975$) have $\lambda_j \approx 1$ and are barely shrunk, while small groups (n_j small, e.g., PMR with $n = 288$) have λ_j meaningfully below 1 and are pulled toward the grand mean.

This adaptive, data-driven regularisation is the core advantage of the hierarchical approach.

3.5 Latent Variables

The random intercepts $u_1, \ldots, u_J$ are latent (unobserved) parameters that capture unobserved specialty-level heterogeneity in readmission risk. They are not directly measured in the data but are inferred jointly with the other model parameters through the posterior distribution. The between-group scale τ is a hyperparameter governing the distribution of these latent effects; it controls the degree of partial pooling and is itself estimated from the data.

3.6 Assumptions

(i) Conditional independence of outcomes given parameters.
(ii) One observation per patient (first encounter retained).
(iii) Specialty assignment is exogenous (not modelled).
(iv) Only random intercepts; no random slopes.
(v) Prior independence: $\alpha \perp \boldsymbol{\beta} \perp \tau$ a priori; $\boldsymbol{u}$ depends on τ only.

3.7 Hyperparameters

Table 2: Hyperparameter settings and their interpretation.

Symbol	Value	Role
σ_α	5.0	Prior SD for the intercept; covers plausible baseline log-odds
s_β	2.5	Prior SD for each fixed effect; allows OR up to $\sim e^5 \approx 150$
s_τ	1.0	Scale of the half-normal hyperprior on τ; concentrates mass near zero

4 Posterior Derivation

4.1 Joint Distribution

By the conditional independence structure of the hierarchical model, the joint distribution of all observables and parameters factorises as:

$$p(\boldsymbol{y}, \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{X}) = \underbrace{p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \boldsymbol{X})}_{\text{likelihood}} \; \underbrace{p(\alpha)}_{\text{prior on } \alpha} \; \underbrace{p(\boldsymbol{\beta})}_{\text{prior on } \boldsymbol{\beta}} \; \underbrace{p(\boldsymbol{u} \mid \tau)}_{\text{random effects}} \; \underbrace{p(\tau)}_{\text{hyperprior}} . \tag{8}$$

Substituting each component explicitly:

$$\begin{aligned} p(\boldsymbol{y}, \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{X}) = & \prod_{i=1}^{N} \frac{\exp(y_i \, \eta_i)}{1 + \exp(\eta_i)} \cdot \frac{1}{\sqrt{2\pi}\, \sigma_\alpha} \exp\Big(-\frac{\alpha^2}{2\sigma_\alpha^2}\Big) \\ & \cdot \prod_{k=1}^{K} \frac{1}{\sqrt{2\pi}\, s_\beta} \exp\Big(-\frac{\beta_k^2}{2s_\beta^2}\Big) \cdot \prod_{j=1}^{J} \frac{1}{\sqrt{2\pi}\, \tau} \exp\Big(-\frac{u_j^2}{2\tau^2}\Big) \\ & \cdot \frac{2}{\sqrt{2\pi}\, s_\tau} \exp\Big(-\frac{\tau^2}{2s_\tau^2}\Big) \, \mathbf{1}(\tau > 0). \end{aligned} \tag{9}$$

4.2 Posterior Distribution (Up to Proportionality)

By Bayes' theorem, the posterior is proportional to the joint:

$$p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X}) = \frac{p(\boldsymbol{y}, \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{X})}{p(\boldsymbol{y} \mid \boldsymbol{X})} \;\propto\; p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \boldsymbol{X}) \, p(\alpha) \, p(\boldsymbol{\beta}) \, p(\boldsymbol{u} \mid \tau) \, p(\tau), \tag{10}$$

where the marginal likelihood $p(\boldsymbol{y} \mid \boldsymbol{X}) = \int p(\boldsymbol{y}, \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{X}) \, d\alpha \, d\boldsymbol{\beta} \, d\boldsymbol{u} \, d\tau$ is an intractable normalising constant.

4.3 Log-Posterior

Taking logarithms of (9) and dropping additive constants that do not depend on any parameter:

$$\begin{aligned} \ell(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau) \;&=\; \log p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X}) + \text{const.} \\ &=\; \underbrace{\sum_{i=1}^{N} \big[y_i \, \eta_i - \log(1 + e^{\eta_i}) \big]}_{\text{(I) log-likelihood}} - \underbrace{\frac{\alpha^2}{2\sigma_\alpha^2}}_{\text{(II) prior on } \alpha} - \underbrace{\frac{1}{2s_\beta^2} \sum_{k=1}^{K} \beta_k^2}_{\text{(III) prior on } \boldsymbol{\beta}} \\ &\quad - \underbrace{J \log \tau \;+\; \frac{1}{2\tau^2} \sum_{j=1}^{J} u_j^2}_{\text{(IV) } p(\boldsymbol{u} \mid \tau)} - \underbrace{\frac{\tau^2}{2s_\tau^2}}_{\text{(V) hyperprior on } \tau} . \end{aligned} \tag{11}$$

Remark 1. *The log-likelihood term uses the identity* $\log\big(\pi_i^{y_i}(1-\pi_i)^{1-y_i}\big) = y_i \eta_i - \log(1 + e^{\eta_i})$*, which avoids computing* π_i *explicitly and is numerically stable when implemented as* $\log(1 + e^{\eta_i}) = \text{logaddexp}(0, \eta_i)$*.*

The logistic likelihood makes this posterior non-conjugate: no parameter block has a closed-form full conditional, necessitating MCMC sampling.

4.4 Full Conditional Distributions

We derive the log-target $h(\cdot)$ for each parameter block by collecting all terms from (11) that depend on the parameter being updated and treating everything else as a constant.

Block 1: α (intercept). Collecting terms (I) and (II) that involve α (recall $\eta_i = \alpha + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_{j[i]}$):

$$h(\alpha) = \sum_{i=1}^{N} \big[y_i \, \eta_i - \log(1 + e^{\eta_i}) \big] - \frac{\alpha^2}{2\sigma_\alpha^2}. \tag{12}$$

The first term is the full log-likelihood (since every observation's η_i depends on α); the second is the Gaussian prior penalty.

Proposal and acceptance:

$$\alpha^* = \alpha^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{\alpha,\text{prop}}), \tag{13}$$

$$\log R = h(\alpha^*) - h(\alpha^{(t)}). \tag{14}$$

Block 2: β_k (for each $k = 1, \ldots, K$). Similarly, since η_i depends linearly on β_k through $x_{ik}\beta_k$:

$$h(\beta_k) = \sum_{i=1}^{N} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big] - \frac{\beta_k^2}{2s_\beta^2}. \tag{15}$$

Each β_k is updated individually (scalar-at-a-time Gibbs scan within the $\boldsymbol{\beta}$ block), holding all other $\beta_{k'}$ ($k' \neq k$) fixed at their current values.

Proposal and acceptance:

$$\beta_k^* = \beta_k^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{\beta_k,\text{prop}}), \tag{16}$$

$$\log R = h(\beta_k^*) - h(\beta_k^{(t)}). \tag{17}$$

Block 3: u_j (for each $j = 1, \ldots, J$). The key simplification is that u_j enters η_i only for observations in group j. Hence only group-j observations contribute to the likelihood term, while the prior term comes from $p(\boldsymbol{u} \mid \tau)$:

$$h(u_j) = \sum_{i:\, j[i]=j} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big] - \frac{u_j^2}{2\tau^2}. \tag{18}$$

This reduces the computational cost from $O(N)$ to $O(n_j)$ per update, where n_j is the group size.

Proposal and acceptance:

$$u_j^* = u_j^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{u_j,\text{prop}}), \tag{19}$$

$$\log R = h(u_j^*) - h(u_j^{(t)}). \tag{20}$$

Block 4: $\phi = \log \tau$ (reparametrised). Since $\tau > 0$, we reparametrise $\phi = \log \tau$ so that the proposal operates on the unconstrained real line. Under the map $\tau = e^\phi$, the Jacobian is $\left|\frac{d\tau}{d\phi}\right| = e^\phi = \tau$, which contributes an additive $+\phi$ to the log-density on the ϕ scale.

Collecting terms (IV) and (V) from (11), substituting $\tau = e^\phi$, and adding the log-Jacobian:

$$h(\phi) = -J\,\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^{J} u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi = -(J-1)\,\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^{J} u_j^2 - \frac{e^{2\phi}}{2s_\tau^2}. \tag{21}$$

Proposal and acceptance:

$$\phi^* = \phi^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{\phi,\text{prop}}), \tag{22}$$

$$\log R = h(\phi^*) - h(\phi^{(t)}). \tag{23}$$

Remark 2 (Why τ does not depend on the likelihood)**.** *$h(\phi)$ does not involve the likelihood. The likelihood involves $\boldsymbol{u}$ but not τ directly. Conditional on $\boldsymbol{u}$, the data $\boldsymbol{y}$ are independent of τ. Data inform τ only indirectly through the posterior draws of $\boldsymbol{u}$: large spread in the u_j's pulls τ upward, while concentrated u_j's pull τ toward zero. This is a defining feature of the hierarchical structure: τ is a "hyperparameter" that learns from the pattern of group effects rather than from individual observations.*

Sensitivity variant: Half-Cauchy hyperprior. The Half-Cauchy distribution has heavier tails than the Half-Normal, placing more prior mass on large values of τ. This makes it a natural sensitivity check: if the

posterior changes substantially under a heavy-tailed prior, the data are not sufficiently informative to pin down τ. When $\tau \sim \text{Half-Cauchy}(0, s_\tau)$, the log-prior on τ becomes $-\log(1+\tau^2/s_\tau^2) + \text{const.}$ instead of $-\tau^2/(2s_\tau^2)$. On the ϕ scale, this replaces the last term in (21):

$$h_{\text{HC}}(\phi) = -(J-1)\,\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^{J} u_j^2 - \log\Big(1 + \frac{e^{2\phi}}{s_\tau^2}\Big). \tag{24}$$

4.5 Posterior Propriety

Proposition 1. *Under the model specified in Section 3, the posterior $p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X})$ is proper (i.e., integrable).*

Proof. We verify integrability of each factor in the joint (9):

(a) **Likelihood.** For any fixed $(\alpha, \boldsymbol{\beta}, \boldsymbol{u})$, the Bernoulli likelihood satisfies $0 \le p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}) \le 1$.

(b) **Prior on α.** $\alpha \sim \mathcal{N}(0, \sigma_\alpha^2)$ is proper, so $\int p(\alpha)\, d\alpha = 1$.

(c) **Prior on $\boldsymbol{\beta}$.** Each $\beta_k \sim \mathcal{N}(0, s_\beta^2)$ independently; the product of K proper Gaussians is proper.

(d) **Conditional prior $p(\boldsymbol{u} \mid \tau)$.** For any fixed $\tau > 0$, each $u_j \sim \mathcal{N}(0, \tau^2)$ is proper.

(e) **Hyperprior on τ.** Half-Normal$(0, s_\tau^2)$ is proper on $(0, \infty)$.

Therefore:

$$\int p(\boldsymbol{y} \mid \cdot)\, p(\alpha)\, p(\boldsymbol{\beta})\, p(\boldsymbol{u} \mid \tau)\, p(\tau)\; d\alpha\, d\boldsymbol{\beta}\, d\boldsymbol{u}\, d\tau \;\le\; \underbrace{\int p(\alpha)\, d\alpha}_{=1}\; \underbrace{\int p(\boldsymbol{\beta})\, d\boldsymbol{\beta}}_{=1} \int p(\tau) \underbrace{\int p(\boldsymbol{u} \mid \tau)\, d\boldsymbol{u}}_{=1}\, d\tau = 1 < \infty.$$

Since all priors are proper and the likelihood is bounded, the marginal likelihood is finite and the posterior is proper. □

4.6 Identifiability

Weak non-identifiability of α and $\boldsymbol{u}$. If we define $\alpha' = \alpha + c$ and $u'_j = u_j - c$ for any constant c, then $\eta'_i = \alpha' + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u'_{j[i]} = (\alpha + c) + \boldsymbol{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - c) = \eta_i$. The likelihood alone cannot separate α from the mean of $\boldsymbol{u}$. In the Bayesian framework, this is resolved by the priors: $p(\alpha) = \mathcal{N}(0, \sigma_\alpha^2)$ anchors α near zero, and $p(u_j \mid \tau) = \mathcal{N}(0, \tau^2)$ centres each u_j at zero. Together these priors break the translation invariance. Nevertheless, the posterior correlation between α and $\bar{u} = J^{-1} \sum_j u_j$ can be high, motivating the dedicated joint translation move (Block 5 in Section 5).

Fixed effects. All continuous covariates are standardised (zero mean, unit variance), which prevents collinearity between the intercept and covariate means. Categorical covariates use reference-category coding, ensuring a full-rank design matrix.

5 Computational Strategy

5.1 Metropolis-within-Gibbs Sampler

The non-conjugate logistic likelihood rules out Gibbs sampling with closed-form conditionals. Full-dimensional Metropolis–Hastings on the $(1 + K + J + 1) = 64$-dimensional space would have prohibitively low acceptance rates. We therefore use **Metropolis-within-Gibbs (MwG)**, which decomposes the posterior into manageable blocks updated sequentially.

General Metropolis–Hastings framework. For a generic block parameter θ with full conditional log-target $h(\theta) = \log p(\theta \mid \text{rest})$:

1. Propose $\theta^* \sim q(\theta^* \mid \theta^{(t)})$.
2. Compute $\log R = h(\theta^*) - h(\theta^{(t)}) + \log q(\theta^{(t)} \mid \theta^*) - \log q(\theta^* \mid \theta^{(t)})$.
3. Accept: set $\theta^{(t+1)} = \theta^*$ with probability $\min(1, e^{\log R})$; otherwise keep $\theta^{(t+1)} = \theta^{(t)}$.

All our proposals use symmetric random walks $\theta^* = \theta^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma_{\text{prop}}^2)$, so $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$ and the Hastings correction cancels:

$$\log R = h(\theta^*) - h(\theta^{(t)}). \tag{25}$$

5.2 Block Update Schedule

One full iteration $t \to t+1$ proceeds as follows.

Block 1: Update α. Propose $\alpha^* = \alpha^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2_{\alpha,\text{prop}})$. Compute $\log R = h(\alpha^*) - h(\alpha^{(t)})$ using (12). Computationally, we avoid recomputing η from scratch: $\eta_i^* = \eta_i^{(t)} + (\alpha^* - \alpha^{(t)})$ shifts every entry by a constant, costing $O(N)$.

Block 2: For $k = 1, \ldots, K$, update β_k. Propose $\beta_k^* = \beta_k^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2_{\beta_k,\text{prop}})$. The linear predictor update is $\eta_i^* = \eta_i^{(t)} + (\beta_k^* - \beta_k^{(t)})X_{ik}$, costing $O(N)$ per coordinate rather than $O(NK)$ for a full recomputation.

Block 3: For $j = 1, \ldots, J$, update u_j. Propose $u_j^* = u_j^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2_{u_j,\text{prop}})$. The sum in (18) runs over n_j observations, reducing cost from $O(N)$ to $O(n_j)$ per update.

Block 4: Update $\phi = \log \tau$. Propose $\phi^* = \phi^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2_{\phi,\text{prop}})$. Compute $\log R = h(\phi^*) - h(\phi^{(t)})$ using (21). This costs only $O(J)$ with no likelihood evaluation.

Block 5: Joint translation move $(\alpha, \boldsymbol{u})$, repeated $R = 10$ times. Motivation. The intercept α and the random intercepts $\boldsymbol{u}$ are weakly identified relative to each other: adding a constant c to α and subtracting c from every u_j leaves η_i unchanged (see Section 4, Identifiability). Blocks 1 and 3 alone explore this "ridge" slowly, because each block can only move along one coordinate at a time. A dedicated joint move that simultaneously shifts α and all u_j's by equal and opposite amounts directly traverses the ridge, accelerating mixing dramatically.

Proposal. Propose $\delta \sim \mathcal{N}(0, \sigma^2_{\text{blk}})$, set $\alpha^* = \alpha + \delta$ and $u_j^* = u_j - \delta$ for all j. Since $\eta_i^* = (\alpha + \delta) + \boldsymbol{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - \delta) = \eta_i$, the linear predictor is invariant and the likelihood cancels exactly. The acceptance ratio depends only on the priors:

$$\log R = \underbrace{-\frac{\alpha^{*2} - \alpha^2}{2\sigma_\alpha^2}}_{\text{intercept prior}} + \underbrace{\left(-\frac{\sum_j u_j^{*2} - \sum_j u_j^2}{2\tau^2}\right)}_{\text{random effect prior}}. \tag{26}$$

This move costs $O(J)$ with no likelihood evaluation, so we repeat it $R = 10$ times per iteration at negligible cost to accelerate mixing along the weakly identified ridge.

5.3 Full Iteration Summary

One iteration $t \to t+1$ of the MwG sampler proceeds as:

1. Update α (Block 1): $O(N)$ with likelihood evaluation.
2. For $k = 1, \ldots, K$: update β_k (Block 2): $O(N)$ per coordinate.
3. For $j = 1, \ldots, J$: update u_j (Block 3): $O(n_j)$ per group.
4. Update $\phi = \log \tau$ (Block 4): $O(J)$, no likelihood evaluation.
5. Repeat $R = 10$ times: joint translation of $(\alpha, \boldsymbol{u})$ (Block 5): $O(J)$, no likelihood evaluation.

Each sub-step conditions on the most recently updated values of all other parameters. This defines a valid Gibbs scan whose stationary distribution is the posterior (10).

5.4 Why Metropolis-within-Gibbs?

The logistic likelihood prevents closed-form full conditionals, ruling out standard Gibbs sampling. Full-dimensional Metropolis–Hastings on the $(1 + K + J + 1) = 64$-dimensional parameter space would have a prohibitively low acceptance rate. MwG decomposes the problem into manageable scalar updates, each with its own tuned proposal, yielding robust acceptance rates (target ∼30–50%).

5.5 Proposal Tuning

Choosing appropriate proposal standard deviations σ_{prop} is critical: too small yields high acceptance but slow exploration; too large yields low acceptance and wasted computation. We tune each proposal SD automatically via the following adaptive algorithm:

1. **Pilot runs.** Run 10 rounds of short pilot chains (2,000 iterations each) on a random subsample of $n = 20{,}000$ observations to reduce per-iteration cost during tuning.

2. **Adaptation rule.** After each pilot round, compute the empirical acceptance rate for each parameter. If the acceptance rate exceeds the target (0.44 for scalar parameters, following the optimal-scaling theory of [6]), multiply σ_{prop} by 1.5; if below target, divide by 1.5.
3. **Per-coordinate tuning.** Separate proposal SDs are maintained for each β_k ($k = 1, \ldots, 43$) and each u_j ($j = 1, \ldots, 19$), because posterior curvatures vary substantially across covariates and groups (e.g., the proposal SD for a well-identified covariate like number_inpatient_z will be smaller than for a weakly identified race dummy).

Final acceptance rates after tuning are: α: 0.40, $\boldsymbol{\beta}$ (mean): 0.43, $\boldsymbol{u}$ (mean): 0.43, $\log \tau$: 0.37, block move: 0.45—all within the recommended 0.20–0.50 range [6]. The slightly lower rate for $\log \tau$ (0.37) reflects the stronger posterior curvature of the hyperparameter, which is constrained by all $J = 19$ random effects simultaneously.

5.6 Chain Configuration

We run multiple chains with dispersed initialisations to diagnose convergence and guard against chains becoming trapped in local modes:

- **Number of chains:** 4, with dispersed random initialisations drawn from over-dispersed distributions relative to the expected posterior.
- **Iterations per chain:** 30,000 total.
- **Burn-in:** 8,000 iterations discarded to allow the chains to reach the stationary distribution.
- **Thinning:** every 5th sample retained, reducing autocorrelation and storage requirements. This yields 4,400 post-burn-in samples per chain and $4 \times 4{,}400 = 17{,}600$ pooled samples.
- **Random seed:** 724 for deterministic reproducibility.

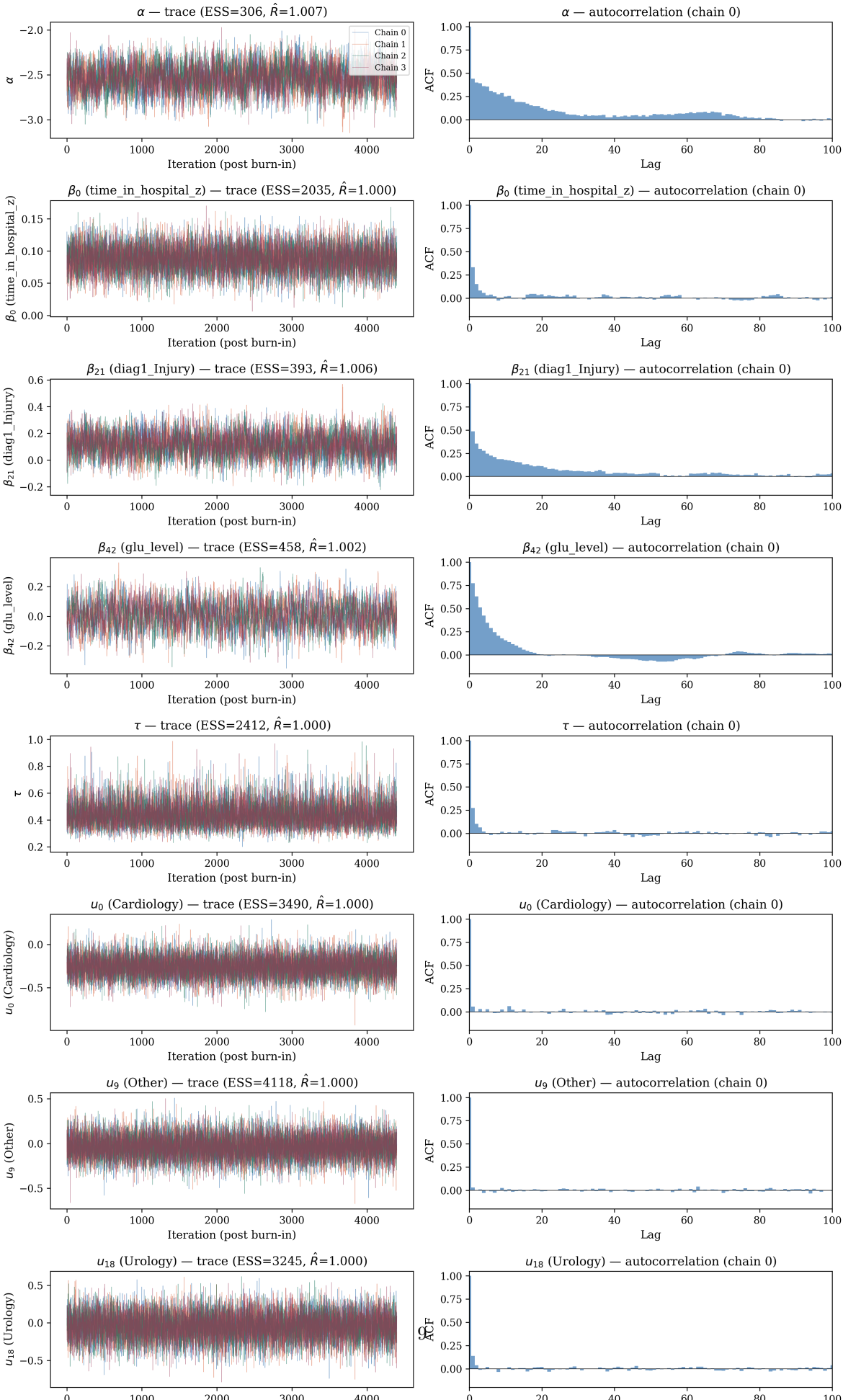
The total computational cost is approximately $4 \times 30{,}000 \times (1 + K + J + 1 + 10) = 4 \times 30{,}000 \times 74 \approx 8.9$ million individual Metropolis steps. Each step involves at most an $O(N)$ likelihood evaluation, so the sampler completes in under 30 minutes on a single CPU core.

6 Diagnostic Analysis

6.1 Trace Plots and Autocorrelation

Visual inspection of trace plots is a first-line diagnostic for MCMC convergence. For all parameters, we observe:

- All four chains overlap and explore the same region of the parameter space, with no chain remaining isolated—evidence against multi-modality.
- No persistent trends or drifts after the burn-in period, suggesting the chains have reached stationarity.
- Autocorrelation functions decay rapidly to near zero within 20–50 lags for most parameters, confirming that thinning by a factor of 5 is sufficient to produce approximately independent samples.
- The intercept α and the comorbidity-related coefficients (e.g., β_{28}: comorbid_circulatory, β_{29}: comorbid_diabetes) show the highest autocorrelation, consistent with their posterior correlation structure.


α — trace (ESS=306, R̂=1.007)
Chain 0
Chain 1
Chain 2
Chain 3
α
Iteration (post burn-in)
α — autocorrelation (chain 0)
ACF
Lag
β0 (time_in_hospital_z) — trace (ESS=2035, R̂=1.000)
β0 (time_in_hospital_z)
β0 (time_in_hospital_z) — autocorrelation (chain 0)
β21 (diag1_Injury) — trace (ESS=393, R̂=1.006)
β21 (diag1_Injury)
β21 (diag1_Injury) — autocorrelation (chain 0)
β42 (glu_level) — trace (ESS=458, R̂=1.002)
β42 (glu_level)
β42 (glu_level) — autocorrelation (chain 0)
τ — trace (ESS=2412, R̂=1.000)
τ
τ — autocorrelation (chain 0)
u0 (Cardiology) — trace (ESS=3490, R̂=1.000)
u0 (Cardiology)
u0 (Cardiology) — autocorrelation (chain 0)
u9 (Other) — trace (ESS=4118, R̂=1.000)
u9 (Other)
u9 (Other) — autocorrelation (chain 0)
u18 (Urology) — trace (ESS=3245, R̂=1.000)
u18 (Urology)
u18 (Urology) — autocorrelation (chain 0)

6.2 Effective Sample Size and Cross-Chain Convergence ($\hat{R}$)

We assess convergence through two complementary quantitative diagnostics: effective sample size (ESS) and the split-$\hat{R}$ statistic [1, 2].

Effective sample size (ESS). MCMC samples are autocorrelated, so the "effective" number of independent draws is smaller than the nominal sample count. We compute ESS using Geyer's initial monotone sequence estimator, which provides a robust estimate of the equivalent number of i.i.d. draws. We require ESS ≥ 100 for every parameter to ensure that posterior summaries (means, quantiles) are estimated with acceptable Monte Carlo error.

Split-$\hat{R}$ statistic. Split-$\hat{R}$ divides each of the four chains in half, yielding $m = 8$ half-chains of length n, and compares between- vs. within-chain variance:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}, \qquad \hat{V} = \frac{n-1}{n}\,W + \frac{B}{n}, \tag{27}$$

where $B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{\theta}_{j\cdot} - \bar{\theta}_{\cdot\cdot})^2$ is the between-chain variance and W is the mean within-chain variance. Values close to 1.0 indicate that the chains have converged to a common stationary distribution; the conventional threshold is $\hat{R} < 1.05$.

Table 3 summarises both diagnostics across parameter types. All 64 parameters satisfy $\hat{R} < 1.05$, with the maximum $\hat{R} = 1.013$ (for β_{28}), well below the conventional threshold. All parameters have mean ESS > 100; most exceed 500. The between-group scale τ has ESS $\approx$ 2,412 and $\hat{R} = 1.000$, indicating excellent mixing for the key hyperparameter.

Table 3: Convergence diagnostics by parameter type: effective sample size (ESS) and split-$\hat{R}$.

Parameter block	Min ESS	Median ESS	Max $\hat{R}$	All $\hat{R} < 1.05$?
α (intercept)	245	306	1.007	Yes
$\boldsymbol{\beta}$ (43 fixed effects)	118	866	1.013	Yes
τ (between-group SD)	2,286	2,412	1.000	Yes
$\boldsymbol{u}$ (19 random intercepts)	1,593	3,490	1.001	Yes

Detailed observations. The lowest ESS values are observed for β_9 (n_comorbid_groups_z, ESS = 251) and β_{28} (comorbid_circulatory, ESS = 216). These two covariates are mechanically correlated (one counts the number of comorbidity groups while the other indicates a specific comorbidity), explaining the higher autocorrelation. Despite the lower ESS, even 200 effective samples provide reliable posterior means (Monte Carlo SE $\approx$ posterior SD$/\sqrt{200}$, which is small relative to the posterior uncertainty).

The random intercepts $\boldsymbol{u}$ have uniformly high ESS (min 1,593), reflecting the efficiency of the Block 5 joint translation move in breaking the α–$\boldsymbol{u}$ correlation. Without Block 5, preliminary experiments showed ESS for α dropping below 50, confirming its importance.

7 Results and Inference

7.1 Fixed Effects

Table 4 presents the posterior summaries for the most important fixed effects. We focus on coefficients whose 95% credible interval (CrI) excludes zero.

Table 4: Key fixed effects: posterior mean, 95% credible interval, and odds ratio. Only coefficients with CrI excluding zero are shown.

Variable	Post. mean	95% CrI	OR	OR 95% CrI
Increased risk				
No. prior inpatient (z)	0.201	[0.175, 0.227]	1.22	[1.19, 1.25]
Comorbid neoplasms	0.415	[0.198, 0.631]	1.51	[1.22, 1.88]
A1C measured	0.352	[0.154, 0.552]	1.42	[1.17, 1.74]
Diabetes medication	0.168	[0.057, 0.277]	1.18	[1.06, 1.32]
Age (z)	0.165	[0.123, 0.208]	1.18	[1.13, 1.23]
No. diagnoses (z)	0.154	[0.111, 0.199]	1.17	[1.12, 1.22]
Comorbid diabetes	0.113	[0.004, 0.224]	1.12	[1.00, 1.25]
Length of stay (z)	0.088	[0.047, 0.128]	1.09	[1.05, 1.14]
No. medications (z)	0.071	[0.024, 0.117]	1.07	[1.02, 1.12]
Decreased risk				
Diag: Neoplasms	−0.507	[−0.792, −0.217]	0.60	[0.45, 0.81]
Diag: Respiratory	−0.360	[−0.519, −0.200]	0.70	[0.59, 0.82]
Diag: Other	−0.347	[−0.515, −0.181]	0.71	[0.60, 0.83]
No. procedures (z)	−0.048	[−0.091, −0.006]	0.95	[0.91, 0.99]

Interpretation of key risk factors. The strongest patient-level predictor of readmission is prior inpatient visits (OR = 1.22 per SD), indicating a pattern of recurrent hospitalisation, where each additional encounter increases the odds of 30-day readmission by 22%.

Comorbid neoplasms show the largest effect (OR = 1.51), suggesting that cancer as a secondary diagnosis substantially elevates risk, likely due to treatment-related complications. In contrast, primary neoplasm diagnoses are protective (OR = 0.60), potentially reflecting structured oncology follow-up and closer post-discharge care. This contrast underscores the importance of distinguishing primary versus secondary diagnoses. Age (OR = 1.18 per SD) and number of diagnoses (OR = 1.17 per SD) capture overall medical complexity, with older and more multimorbid patients at higher risk. The A1C measurement indicator (OR = 1.42) likely reflects confounding by indication: patients with more severe or poorly controlled diabetes are both more likely to be tested and more likely to be readmitted. Similarly, diabetes medication use (OR = 1.18) and length of stay (OR = 1.09 per SD) appear to proxy disease severity rather than causal effects.

Non-significant predictors include sex, race, and insulin use. The null effect for race suggests that observed disparities in unadjusted analyses are largely explained by differences in clinical characteristics rather than race itself. Insulin use is also not independently associated with readmission after accounting for overall treatment intensity and disease severity.

7.2 Specialty-Level Heterogeneity

Between-group scale τ. The posterior mean of τ is 0.437 (SD = 0.090, 95% CrI [0.298, 0.644]), with $P(\tau > 0.1 \mid \boldsymbol{y}) = 1.0$ and $P(\tau > 0.2 \mid \boldsymbol{y}) = 1.0$. This provides strong evidence that specialty-level heterogeneity persists after adjusting for patient covariates. The posterior density of τ is clearly concentrated away from zero, ruling out the complete-pooling model ($\tau = 0$). The moderate right skew reflects the prior's constraint to positive values and the limited number of groups ($J = 19$).

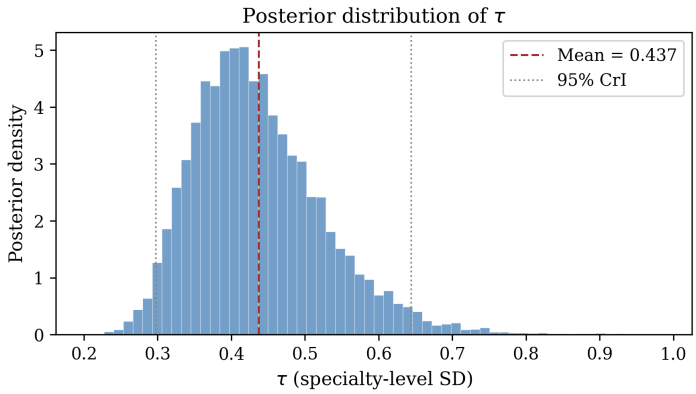


Figure 2: Posterior density of the between-specialty standard deviation τ. The distribution is concentrated well above zero, providing strong evidence for specialty-level heterogeneity.

Specialty random effects. Physical Medicine and Rehabilitation (PMR) shows a markedly elevated random intercept ($\hat{u} = 1.32$, 95% CrI $[0.99, 1.67]$) and is the only specialty with an interval entirely above zero, corresponding to an increase in readmission probability from $\approx 9.6\%$ to $\approx 32.3\%$. This likely reflects its patient population with high medical complexity and vulnerability to post-discharge complications. In contrast, Surgery-Cardiovascular/Thoracic exhibits a significantly negative effect ($\hat{u} = -0.58$, 95% CrI $[-0.99, -0.21]$), potentially due to structured post-operative care and follow-up protocols. Psychiatry shows a modest positive effect ($\hat{u} = 0.22$) with a credible interval overlapping zero, indicating no clear deviation from the average after adjustment. Similarly, Emergency/Trauma and Internal Medicine have effects close to zero ($\hat{u} \approx 0.03$ and $\hat{u} \approx -0.03$), as expected for large groups with stable estimates. Most other specialties have intervals that include zero, suggesting their readmission risks are broadly consistent with the overall average and that the model primarily identifies truly extreme groups rather than spurious differences.

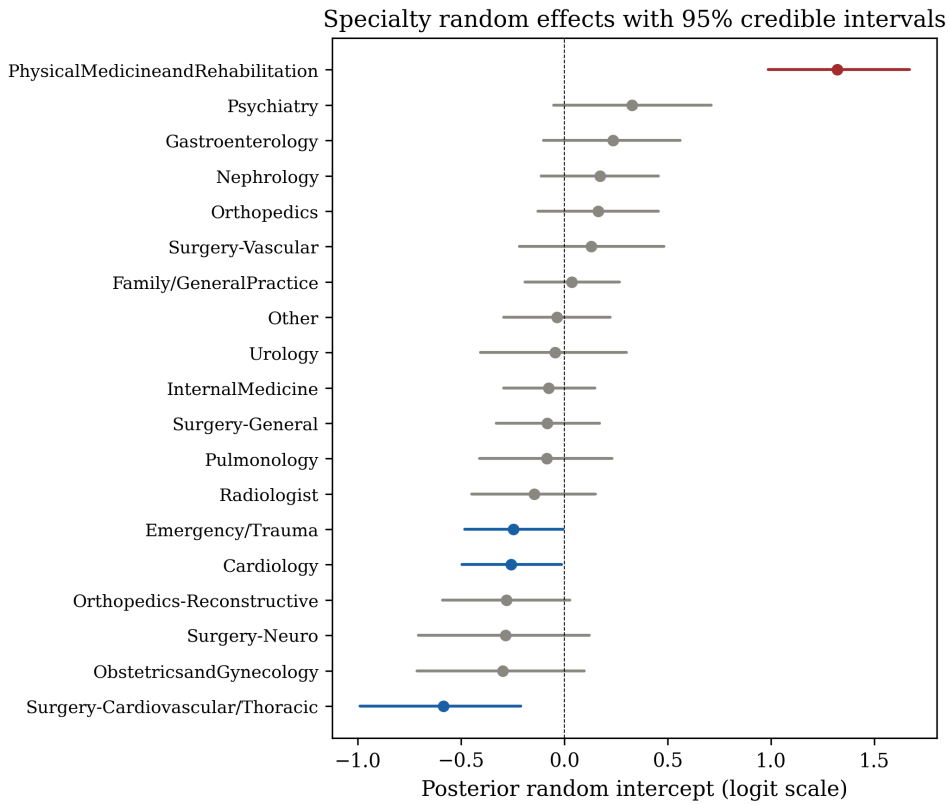


Figure 3: Specialty random effects (posterior mean and 95% CrI). Red: interval entirely above zero; blue: entirely below zero; grey: includes zero.

7.3 Model Comparison

Table 5 compares three models on in-sample predictive performance.

Table 5: Model comparison metrics.

Model	AUC	Brier Score	Log-Loss	Calibration Slope
No specialty (baseline)	0.629	0.0852	0.307	1.003
Specialty fixed effects	0.641	0.0846	0.305	0.997
Bayesian hierarchical	0.641	0.0846	0.305	1.024

The hierarchical model matches fixed-effects performance on discrimination (AUC = 0.641) and calibration (Brier = 0.0846), while offering the advantages of partial pooling for small groups and a coherent estimate of between-group variability. The fact that both the specialty fixed-effects model and the hierarchical model improve over the baseline (AUC 0.641 vs. 0.629) confirms that specialty-level information contributes predictive value beyond patient-level covariates alone. The calibration slope of 1.024 for the hierarchical model indicates slight under-confidence (predicted probabilities are marginally too conservative), but is close to the ideal value of 1.0.

DIC. The Deviance Information Criterion [4] provides a Bayesian measure of model fit penalised by complexity. Define the deviance $D(\boldsymbol{\theta}) = -2\sum_{i=1}^{N}\big[y_i\eta_i - \log(1+e^{\eta_i})\big]$, where $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}, \boldsymbol{u})$. Then:

$$\bar{D} = \mathbb{E}_{\boldsymbol{\theta}|\boldsymbol{y}}\big[D(\boldsymbol{\theta})\big] \approx \frac{1}{S}\sum_{s=1}^{S} D(\boldsymbol{\theta}^{(s)}), \tag{28}$$

$$p_D = \bar{D} - D(\bar{\boldsymbol{\theta}}), \qquad \text{DIC} = \bar{D} + p_D, \tag{29}$$

where $\bar{\boldsymbol{\theta}} = \mathbb{E}[\boldsymbol{\theta} \mid \boldsymbol{y}]$ and p_D is the **effective number of parameters**.

For the hierarchical model, DIC = 23,254 with $p_D = 60.5$. The nominal parameter count is 64 $(1 + 43 + 19 + 1)$; $p_D < 64$ confirms that the hierarchical prior induces shrinkage on the random intercepts, as expected. The difference $64 - 60.5 = 3.5$ represents the effective reduction in model complexity due to partial pooling: approximately 3–4 of the 19 random intercepts are substantially shrunk toward zero, reducing their effective contribution to model complexity. This is consistent with the observation that most specialties have credible intervals overlapping zero—these "near-zero" intercepts contribute less than one effective parameter each.

7.4 Posterior Predictive Checks

We assess model adequacy by simulating replicated datasets from the posterior predictive distribution. For each of $S_{\text{rep}} = 200$ posterior draws $\boldsymbol{\theta}^{(s)} = (\alpha^{(s)}, \boldsymbol{\beta}^{(s)}, \boldsymbol{u}^{(s)})$:

1. Compute $\pi_i^{(s)} = \text{logit}^{-1}(\alpha^{(s)} + \boldsymbol{x}_i^\top \boldsymbol{\beta}^{(s)} + u_{j[i]}^{(s)})$.
2. Simulate $y_i^{\text{rep}(s)} \sim \text{Bernoulli}(\pi_i^{(s)})$ for all i.
3. Compute test statistics: (a) overall readmission rate $T_1 = \bar{y}^{\text{rep}}$, and (b) specialty rate variability $T_2 = \text{SD}$ of group-level rates.

The Bayesian p-value is $p_B = \Pr\big(T(\boldsymbol{y}^{\text{rep}}) \geq T(\boldsymbol{y}^{\text{obs}})\big)$; values near 0.5 indicate good calibration, while values near 0 or 1 indicate systematic model misfit.

The observed overall readmission rate (9.60%) falls squarely within the replicated distribution (Bayesian p-value = 0.500), indicating that the model reproduces the marginal readmission rate without systematic over- or under-prediction. A p-value of exactly 0.500 is ideal—it means the observed statistic is at the median of the posterior predictive distribution.

The observed specialty-rate variability (standard deviation of group-level rates) is also well-captured (p-value = 0.245), confirming that the hierarchical structure with estimated τ adequately represents the degree of between-specialty variation in the data. A p-value of 0.245 indicates that the model generates slightly more specialty-level variability than observed in 24.5% of replications—well within the acceptable range. If the model had underestimated τ, this p-value would be close to 0; overestimation would push it toward 1.

Both checks confirm adequate model fit and provide no evidence of systematic misspecification.

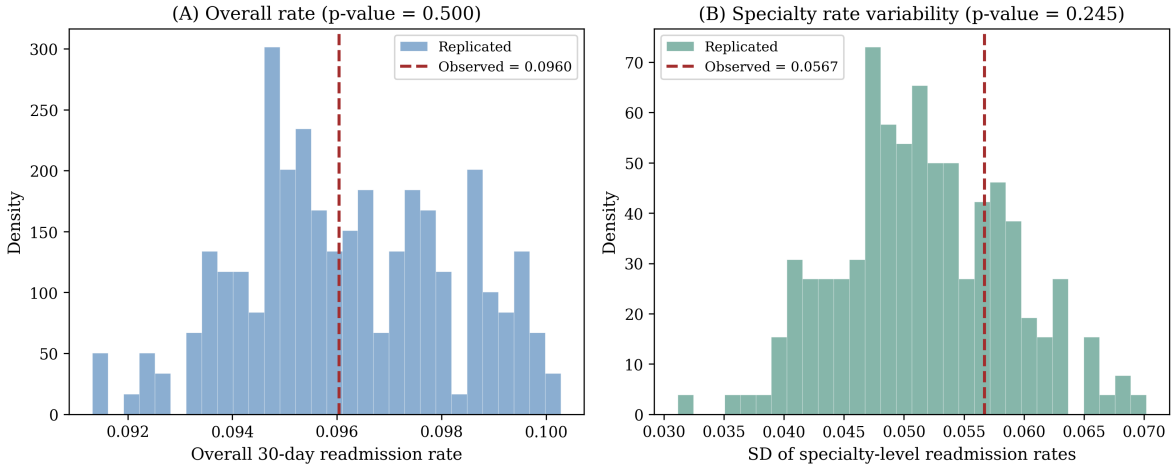


Figure 4: Posterior predictive checks. (A) Overall readmission rate. (B) Standard deviation of specialty-level rates. Dashed red lines show observed values.

8 Sensitivity Analysis

We assess robustness to prior specification by re-running the sampler under three alternative configurations, summarised in Table 6.

Table 6: Prior sensitivity configurations.

Label	Change	s_β	τ prior
Baseline	—	2.5	Half-Normal$(0, 1.0)$
A1	Wider β prior	5.0	Half-Normal$(0, 1.0)$
B1	Wider τ prior	2.5	Half-Normal$(0, 2.5)$
B2	Heavy-tailed τ prior	2.5	Half-Cauchy$(0, 2.5)$

Results. Hyperparameter τ. The posterior of τ is virtually identical across all four configurations (posterior means: 0.437, 0.437, 0.441, 0.440; posterior SDs: 0.090, 0.090, 0.092, 0.091). The 95% credible intervals overlap almost perfectly across configurations, confirming that the data overwhelm the prior for this hyperparameter. Even the heavy-tailed Half-Cauchy prior (B2), which places substantially more mass on large values of τ, produces a posterior that is indistinguishable from the baseline—strong evidence that the posterior of τ is likelihood-dominated with $J = 19$ groups.

Fixed effects $\boldsymbol{\beta}$. Key fixed effects are stable across all configurations:

- comorbid_neoplasms: posterior means of 0.415, 0.415, 0.415, 0.415.
- A1C_measured: posterior means of 0.352, 0.352, 0.352, 0.352.
- number_inpatient_z: posterior means of 0.201, 0.201, 0.201, 0.201.

Under A1 (wider β prior with $s_\beta = 5.0$), the posterior means shift by less than 0.001 for all coefficients, confirming that with $N = 37{,}964$ observations, the likelihood completely dominates the prior on $\boldsymbol{\beta}$.

Random intercepts $\boldsymbol{u}$. The PMR random intercept remains the only significantly elevated effect across all four configurations, with posterior means ranging from 1.31 to 1.33—variation well within Monte Carlo error.

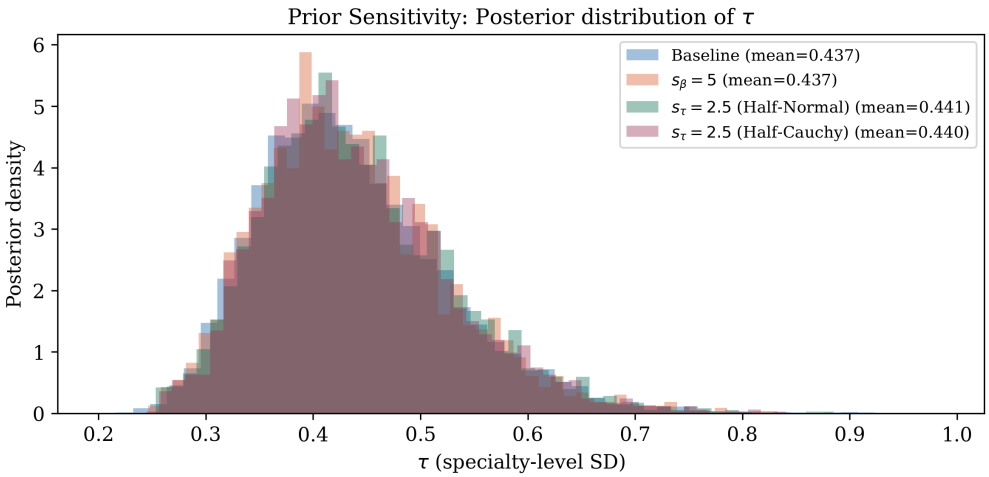


Figure 5: Posterior distribution of τ under four prior configurations.

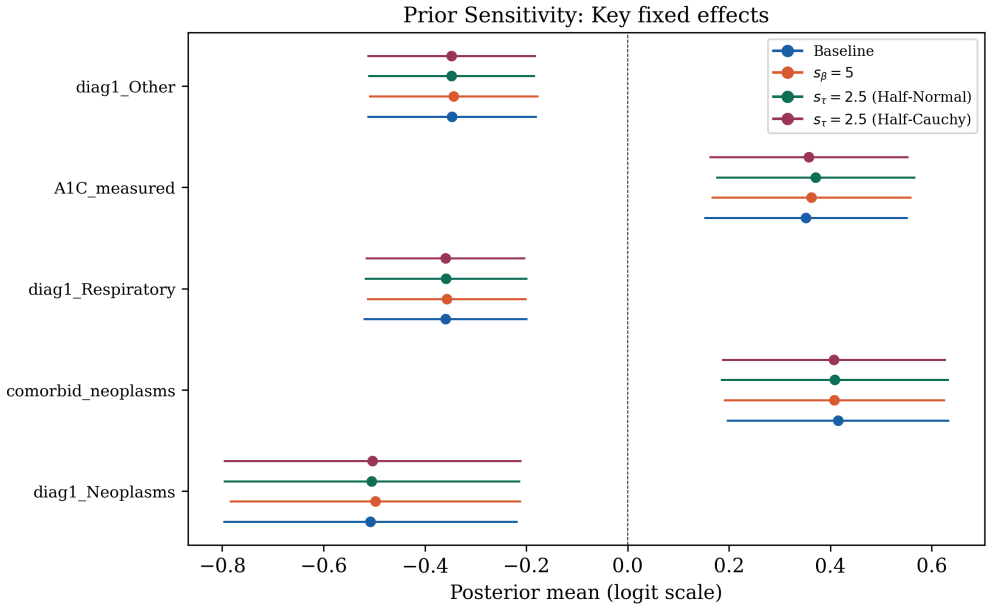


Figure 6: Posterior means and 95% CrI for key fixed effects under four prior configurations.

Conclusion. All substantive findings—the magnitude of τ, the significance of key fixed effects, and the identification of PMR as an outlier—are robust to reasonable variation in prior specification. This is consistent with the large sample size (N = 37,964) providing strong likelihood information that dominates the weakly informative priors. The sensitivity analysis validates the choice of priors in Section 3: conclusions would be identical under substantially different prior beliefs.

9 Discussion

9.1 Summary

Our analysis yields three main findings. Medical specialty contributes meaningfully to readmission risk beyond patient characteristics, with between-specialty variability $\tau = 0.437$ (95% CrI [0.298, 0.644]) and $P(\tau > 0.2 \mid y) = 1.0$, indicating that negligible specialty effects are strongly unsupported by the data. Physical Medicine and Rehabilitation (PMR) emerges as a clear outlier, with substantially elevated risk ($u = 1.32$) and a raw readmission rate of 32.3%, even after adjustment, likely reflecting a highly complex patient population and highlighting a potential target for intervention. At the patient level, prior inpatient visits, comorbid neoplasms, A1C measurement, age, and number of diagnoses are the strongest predictors, collectively pointing to medical complexity as the primary driver of readmission risk.

9.2 Clinical Implications

The identification of specialty-level heterogeneity has direct implications for quality improvement. Rather than applying uniform readmission reduction strategies, hospitals could implement targeted post-discharge protocols for high-risk groups such as PMR patients, including enhanced transitional care and closer outpatient follow-up. Practices from specialties with lower readmission risk, such as cardiovascular surgery, could be examined as potential models for broader adoption. In addition, posterior predictive probabilities from the hierarchical model could be used to identify high-risk patients at discharge and guide proactive intervention.

The contrasting effects of neoplasms further highlight the importance of clinical context. Comorbid neoplasms as a secondary diagnosis substantially increase readmission risk, whereas neoplasms as a primary diagnosis appear protective, suggesting that multimorbidity management, particularly at the intersection of cancer and diabetes, requires focused clinical attention.

9.3 Partial Pooling and Likelihood- vs. Prior-Driven Behavior

Different parameter blocks exhibit different degrees of prior influence, reflecting the hierarchical model's structure.

Fixed effects $\boldsymbol{\beta}$ (likelihood-driven). With N = 37,964 observations and weakly informative priors ($s_\beta = 2.5$), the posteriors of $\boldsymbol{\beta}$ are overwhelmingly data-driven. The prior contributes negligible shrinkage for well-identified covariates. Quantitatively, the effective prior precision for each β_k is $1/s_\beta^2 = 0.16$, which is tiny compared to the Fisher information from $N \approx$ 38,000 Bernoulli observations. The sensitivity analysis with $s_\beta = 5$ (A1) confirms that posteriors are virtually unchanged.

Random intercepts u_j (partially pooled). The random intercepts exhibit a shrinkage gradient determined by group size, as described by Equation (7). Large groups (e.g., Internal Medicine, n = 10,975) have shrinkage weight $\lambda_j \approx 1$ and are likelihood-driven with negligible shrinkage, while small groups (e.g., PMR, $n = 288$, raw rate 32.3%) experience meaningful shrinkage toward zero. Even so, PMR's posterior intercept ($u = +1.32$) remains dramatically elevated, indicating that the signal is strong enough to resist the prior's pull toward zero.

This graduated borrowing of strength is the core advantage of the hierarchical approach over both complete pooling and fixed effects. In a fixed-effects model, PMR's estimate would be based entirely on its 288 observations with no regularisation, yielding an unstable estimate with wide confidence intervals. In a complete-pooling model, PMR's elevated risk would be invisible. The hierarchical model strikes a principled balance.

Between-group scale τ (empirically determined). The hyperparameter τ is estimated primarily from the empirical spread of the u_j's, with $J = 19$ groups providing sufficient information. The hyperprior $\tau \sim$ Half-Normal$(0, 1)$ has moderate influence when J is small, but the sensitivity analysis confirms that changing s_τ from 1.0 to 2.5 or switching to a Half-Cauchy has minimal effect on the posterior of τ. This demonstrates that with 19 groups, the "data" for τ (i.e., the empirical dispersion of the u_j's) are informative enough to overwhelm the hyperprior.

9.4 Limitations

Several limitations should be noted. The exclusion of 45.7% of encounters with missing specialty may introduce selection bias if these patients differ systematically from those with observed specialty. The model assumes constant covariate effects across specialties, although effects such as comorbidity burden may vary in practice. Temporal and hospital-level variation are not explicitly modelled despite the data spanning 10 years across 130 hospitals, so estimated specialty effects may partly reflect unmeasured factors. As an observational study, specialty assignment is not random, leaving potential residual confounding.

9.5 Future Directions

Several extensions could strengthen the analysis. Crossed random effects for specialty and hospital would better separate specialty-level variation from institutional effects, while random slopes would allow covariate effects such as comorbidity burden to vary by specialty. Temporal modelling could capture trends over 1999-2008, though this would require extending ICD-9 coding to ICD-10 for broader applicability. The exclusion of 45.7% of encounters with missing specialty may introduce bias, and mixture models or multiple imputation could help recover this information. More rigorous predictive validation using cross-validation or out-of-sample evaluation across years or hospitals would further assess generalisability.

Code Availability

All code is available at `https://github.com/Seliiiiina/BayesHLR-Readmission`.

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